

 ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)		Orifice Plates and Flanges INSTRUMENTATION DATA SHEET		Document No. B903-073-YF-DS-3100		
PROJECT UNIT SUPPLIER		CLIENT JOB NO. REF. NO.		Rev. No. Page		
EPCM for Offsite Pipelines and Inlet Receiving Tanks Projects for MRPL Refinery Complex IRT system		MRPL - MANGALORE REFINERY		073		
		B903		UNIT NO		
		B903-000-YK-MR-2090				
GENERAL	1	Tag Number		073-FE-1301	Rev No: 0	
	2	Service		EFFLUENT WASTE WATER		
	3	Line No.		8"-WW-73-1311-A9A		
	4	Line Size	Schedule	8"	STD	
	5	Line ID	OD	mm	202.7	219.1
	6	Type of taps		Flange		
FLUID DATA	Condition / Case					
	7	Fluid		EFFLUENT WASTE WATER		
	8	Fluid State		Liquid		
	9	Maximum Flow	m³/h	125		
	10	Normal Flow	m³/h	100		
	11	Minimum Flow	m³/h	50		
	12	Operating Inlet Pressure	kg/cm²g	3.1		
	13	Operating Temperature	°C	ambient		
	14	Design Temperature	°C	65		
	15	Relative Density at 15 deg. C				
	16	Relative Density at Oper. Temp.		0.99 to 1		
	17	Steam Density	kg/m³			
	18	Degree of Super Heat	°C			
	19	Compressibility Factor				
	20	Relative Molecular Mass				
	21	Ratio of Specific Heats				
	22	Operating Viscosity	CST	01 to 10		
	23	Base Pressure	mmH2O			
	24	Base Temperature	°C			
	METER	25	Differential Range	mmH2O	2500	
		26	Upper Range Value	m³/h	150	
	PLATE	27	Type		Conc. Square Edged	
28		Basis of Sizing		ISO 5167		
29		Plate Material		SS 316		
30		Plate Holder	Plate Holder Material	Not Required		
31		Ring Type (for RTJ flanges)				
32		Ring Material (for RTJ flanges)				
33		Beta=d/D		0.538		
34		Radius r Quadrant	mm			
35		Orifice Bore Diameter	mm	109		
36		Vent/Drain Hole Requirement		Vent		
37		Vent/Drain Hole Dia.	mm	3.97		
ORIFICE FLANGES		38	Centre Distance, R	mm		
	39	Plate Thickness, W	mm	6.35		
	40	Reynolds Number		174421		
	41	Flanges		Required		
	42	Type of Flange		Weld Neck		
	43	Tap Size	mm	0.5		
	44	No. of Taps per flange		Two		
	45	Flange Material		ASTM A 105		
	46	Size & Rating		8"	300#	
	47	Facing & Finish		RF 125 AARH		
CERTIFICATION	48	Gasket thickness	mm			
	49	Gasket Material		SP.WND SS316+GRAFIL+ I RING		
	50	Stud Bolt Material		A193 GR.B7		
	51	Nut Material		A194 GR.2H		
OPTIONS	52	Statutory		NA		
	53	Others		NA		
		54	Manufacturers	*		
		55	Model No.	*		
NOTES 1. Orifice assembly must include supply of Plate, Flanges, Studs-Bolts, Nuts, Jack screws, Gasket & Plugs. 2. Gasket Thickness: As per B-16.20- ANSI B16.5. 3. Allowable pressure drop: 0.3 Kg/cm2g.						
[] Deviation		[] No Deviation		Vendor Seal		
0		10-Feb-2025		Issued For Bids		
Rev. No.		Date		Purpose		
Prepared By		Checked By		Approved By		

 ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)		Orifice Plates and Flanges INSTRUMENTATION DATA SHEET		Document No. B903-073-YF-DS-3100		
PROJECT UNIT SUPPLIER		CLIENT JOB NO. REF. NO.		Rev. No. Page		
EPCM for Offsite Pipelines and Inlet Receiving Tanks Projects for MRPL Refinery Complex IRT system		MRPL - MANGALORE REFINERY		073		
		B903		UNIT NO		
		B903-000-YK-MR-2090				
GENERAL	1	Tag Number	073-FE-1302		Rev No: 0	
	2	Service	EFFLUENT WASTE WATER			
	3	Line No.	12"-WWW-73-1312-A9A			
	4	Line Size Schedule	12"		STD	
	5	Line ID OD	mm	304.8	323.9	
	6	Type of taps	Flange			
FLUID DATA	Condition / Case					
	7	Fluid	EFFLUENT WASTE WATER			
	8	Fluid State	Liquid			
	9	Maximum Flow	m³/h	300		
	10	Normal Flow	m³/h	300		
	11	Minimum Flow	m³/h	150		
	12	Operating Inlet Pressure	kg/cm²g	3.1		
	13	Operating Temperature	°C	ambient		
	14	Design Temperature	°C	65		
	15	Relative Density at 15 deg. C				
	16	Relative Density at Oper. Temp.	0.99 to 1			
	17	Steam Density	kg/m³			
	18	Degree of Super Heat	°C			
	19	Compressibility Factor				
	20	Relative Molecular Mass				
	21	Ratio of Specific Heats				
	22	Operating Viscosity	CST	01 to 10		
	23	Base Pressure	mmH2O			
	24	Base Temperature	°C			
	METER	25	Differential Range	mmH2O	2500	
		26	Upper Range Value	m³/h	350	
	PLATE	27	Type	Conc. Square Edged		
28		Basis of Sizing	ISO 5167			
29		Plate Material	SS 316			
30		Plate Holder Plate Holder Material	Not Required			
31		Ring Type (for RTJ flanges)				
32		Ring Material (for RTJ flanges)				
33		Beta=d/D	0.532			
34		Radius r Quadrant	mm			
35		Orifice Bore Diameter	mm	162.1		
36		Vent/Drain Hole Requirement	Vent			
37		Vent/Drain Hole Dia.	mm	5.56		
ORIFICE FLANGES		38	Centre Distance, R	mm		
	39	Plate Thickness, W	mm	6.35		
	40	Reynolds Number	348014			
	41	Flanges	Required			
	42	Type of Flange	Weld Neck			
	43	Tap Size	mm	0.5		
	44	No. of Taps per flange	Two			
	45	Flange Material	ASTM A 105			
	46	Size & Rating	12"	300#		
	47	Facing & Finish	RF 125 AARH			
CERTIFICATION	48	Gasket thickness	mm			
	49	Gasket Material	SP.WND SS316+GRAFIL+ I RING			
	50	Stud Bolt Material	A193 GR.B7			
	51	Nut Material	A194 GR.2H			
OPTIONS	52	Statutory	NA			
	53	Others	NA			
		54	Manufacturers	*		
		55	Model No.	*		
NOTES 1. Orifice assembly must include supply of Plate, Flanges, Studs-Bolts, Nuts, Jack screws, Gasket & Plugs. 2. Gasket Thickness: As per B-16.20- ANSI B16.5. 3. Allowable pressure drop: 0.3 Kg/cm2g.						
[] Deviation		[] No Deviation		Vendor Seal		
0		10-Feb-2025		Issued For Bids		
Rev. No.		Date		Purpose		
Prepared By		Checked By		Approved By		

 ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)		Orifice Plates and Flanges INSTRUMENTATION DATA SHEET		Document No. B903-073-YF-DS-3100		
		Rev. No. 0				
		Page 3		of 5		
PROJECT	EPCM for Offsite Pipelines and Inlet Receiving Tanks Projects for MRPL Refinery Complex		CLIENT	MRPL - MANGALORE REFINERY		
UNIT	IRT system		JOB NO.	B903	UNIT NO. 073	
SUPPLIER			REF. NO.	B903-000-YK-MR-2090		
GENERAL	1	Tag Number		073-FE-1303	Rev No: 0	
	2	Service		EFFLUENT WASTE WATER		
	3	Line No.		12"-WWW-73-1313-A9A		
	4	Line Size Schedule		12"	STD	
	5	Line ID OD	mm	304.8	323.9	
	6	Type of taps		Flange		
FLUID DATA	Condition / Case					
	7	Fluid		EFFLUENT WASTE WATER		
	8	Fluid State		Liquid		
	9	Maximum Flow	m³/h	325		
	10	Normal Flow	m³/h	300		
	11	Minimum Flow	m³/h	150		
	12	Operating Inlet Pressure	kg/cm²g	3.1		
	13	Operating Temperature	°C	ambient		
	14	Design Temperature	°C	65		
	15	Relative Density at 15 deg. C				
	16	Relative Density at Oper. Temp.		0.99 to 1		
	17	Steam Density	kg/m³			
	18	Degree of Super Heat	°C			
	19	Compressibility Factor				
	20	Relative Molecular Mass				
	21	Ratio of Specific Heats				
	22	Operating Viscosity	CST	01 to 10		
	23	Base Pressure	mmH2O			
	24	Base Temperature	°C			
	METER	25	Differential Range	mmH2O	2500	
		26	Upper Range Value	m³/h	350	
	PLATE	27	Type		Conc. Square Edged	
28		Basis of Sizing		ISO 5167		
29		Plate Material		SS 316		
30		Plate Holder Plate Holder Material		Not Required		
31		Ring Type (for RTJ flanges)				
32		Ring Material (for RTJ flanges)				
33		Beta=d/D		0.552		
34		Radius r Quadrant	mm			
35		Orifice Bore Diameter	mm	168.8		
36		Vent/Drain Hole Requirement		Vent		
37		Vent/Drain Hole Dia.	mm	5.56		
ORIFICE FLANGES		38	Centre Distance, R	mm		
	39	Plate Thickness, W	mm	6.35		
	40	Reynolds Number		348014		
	41	Flanges		Required		
	42	Type of Flange		Weld Neck		
	43	Tap Size	mm	0.5		
	44	No. of Taps per flange		Two		
	45	Flange Material		ASTM A 105		
	46	Size & Rating		12"	300#	
	47	Facing & Finish		RF 125 AARH		
CERTIFICATION	48	Gasket thickness	mm			
	49	Gasket Material		SP.WND SS316+GRAFIL+ I RING		
	50	Stud Bolt Material		A193 GR.B7		
	51	Nut Material		A194 GR.2H		
OPTIONS	52	Statutory		NA		
	53	Others		NA		
	54	Manufacturers		*		
	55	Model No.		*		
NOTES 1. Orifice assembly must include supply of Plate, Flanges, Studs-Bolts, Nuts, Jack screws, Gasket & Plugs. 2. Gasket Thickness: As per B-16.20- ANSI B16.5. 3. Allowable pressure drop: 0.3 Kg/cm2g.						
[] Deviation		[] No Deviation		Vendor Seal		
0		10-Feb-2025		Issued For Bids		
Rev. No.	Date	Purpose		Prepared By	Checked By	
				SKG	SM	
					Approved By	

 ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)		Orifice Plates and Flanges INSTRUMENTATION DATA SHEET		Document No. B903-073-YF-DS-3100		
				Rev. No. 0		
				Page 4	of 5	
PROJECT	EPCM for Offsite Pipelines and Inlet Receiving Tanks Projects for MRPL Refinery Complex		CLIENT	MRPL - MANGALORE REFINERY		
UNIT	IRT system		JOB NO.	B903	UNIT NO. 073	
SUPPLIER			REF. NO.	B903-000-YK-MR-2090		
GENERAL	1	Tag Number		073-FE-1304	Rev No: 0	
	2	Service		EFFLUENT WASTE WATER		
	3	Line No.		14"-WWW-73-1316-A9A		
	4	Line Size Schedule		14"	STD	
	5	Line ID OD	mm	336.6	355.6	
	6	Type of taps		Flange		
FLUID DATA	Condition / Case					
	7	Fluid		EFFLUENT WASTE WATER		
	8	Fluid State		Liquid		
	9	Maximum Flow	m³/h	325		
	10	Normal Flow	m³/h	300		
	11	Minimum Flow	m³/h	150		
	12	Operating Inlet Pressure	kg/cm²g	3.1		
	13	Operating Temperature	°C	ambient		
	14	Design Temperature	°C	65		
	15	Relative Density at 15 deg. C				
	16	Relative Density at Oper. Temp.		0.99 to 1		
	17	Steam Density	kg/m³			
	18	Degree of Super Heat	°C			
	19	Compressibility Factor				
	20	Relative Molecular Mass				
	21	Ratio of Specific Heats				
	22	Operating Viscosity	CST	01 to 10		
	23	Base Pressure	mmH2O			
	24	Base Temperature	°C			
	METER	25	Differential Range	mmH2O	2500	
		26	Upper Range Value	m³/h	350	
	PLATE	27	Type		Conc. Square Edged	
28		Basis of Sizing		ISO 5167		
29		Plate Material		SS 316		
30		Plate Holder Plate Holder Material		Not Required		
31		Ring Type (for RTJ flanges)				
32		Ring Material (for RTJ flanges)				
33		Beta=d/D		0.504		
34		Radius r Quadrant	mm			
35		Orifice Bore Diameter	mm	169.6		
36		Vent/Drain Hole Requirement		Vent		
37		Vent/Drain Hole Dia.	mm	5.57		
ORIFICE FLANGES		38	Centre Distance, R	mm		
	39	Plate Thickness, W	mm	6.35		
	40	Reynolds Number		523263		
	41	Flanges		Required		
	42	Type of Flange		Weld Neck		
	43	Tap Size	mm	0.5		
	44	No. of Taps per flange		Two		
	45	Flange Material		ASTM A 105		
	46	Size & Rating		14"	300#	
	47	Facing & Finish		RF 125 AARH		
CERTIFICATION	48	Gasket thickness	mm			
	49	Gasket Material		SP.WND SS316+GRAFIL+ I RING		
	50	Stud Bolt Material		A193 GR.B7		
	51	Nut Material		A194 GR.2H		
OPTIONS	52	Statutory		NA		
	53	Others		NA		
	54	Manufacturers		*		
	55	Model No.		*		
NOTES 1. Orifice assembly must include supply of Plate, Flanges, Studs-Bolts, Nuts, Jack screws, Gasket & Plugs. 2. Gasket Thickness: As per B-16.20- ANSI B16.5. 3. Allowable pressure drop: 0.3 Kg/cm2g.						
[] Deviation		[] No Deviation		Vendor Seal		
0		10-Feb-2025		Issued For Bids		
Rev. No.		Date		Purpose		
Prepared By		Checked By		Approved By		

 ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)		Orifice Plates and Flanges INSTRUMENTATION DATA SHEET		Document No. B903-073-YF-DS-3100		
PROJECT UNIT SUPPLIER		CLIENT JOB NO. REF. NO.		Rev. No. Page		
EPCM for Offsite Pipelines and Inlet Receiving Tanks Projects for MRPL Refinery Complex IRT system		MRPL - MANGALORE REFINERY		0		
		B903		UNIT NO		
		B903-000-YK-MR-2090		073		
GENERAL	1	Tag Number	073-FE-1305		Rev No: 0	
	2	Service	EFFLUENT WASTE WATER			
	3	Line No.	18"-WWW-73-1310-A9A			
	4	Line Size Schedule	18"		STD	
	5	Line ID OD	mm	438.2	457.2	
	6	Type of taps	D-D/2			
FLUID DATA	Condition / Case					
	7	Fluid	EFFLUENT WASTE WATER			
	8	Fluid State	Liquid			
	9	Maximum Flow	m³/h	975		
	10	Normal Flow	m³/h	650		
	11	Minimum Flow	m³/h	325		
	12	Operating Inlet Pressure	kg/cm²g	3.4		
	13	Operating Temperature	°C	ambient		
	14	Design Temperature	°C	65		
	15	Relative Density at 15 deg. C				
	16	Relative Density at Oper. Temp.	0.99 to 1			
	17	Steam Density	kg/m³			
	18	Degree of Super Heat	°C			
	19	Compressibility Factor				
	20	Relative Molecular Mass				
	21	Ratio of Specific Heats				
	22	Operating Viscosity	CST	01 to 10		
	23	Base Pressure	mmH2O			
	24	Base Temperature	°C			
	METER	25	Differential Range	mmH2O	2500	
		26	Upper Range Value	m³/h	1100	
	PLATE	27	Type	Conc. Square Edged		
28		Basis of Sizing	ISO 5167			
29		Plate Material	SS 316			
30		Plate Holder Plate Holder Material	Not Required			
31		Ring Type (for RTJ flanges)				
32		Ring Material (for RTJ flanges)				
33		Beta=d/D	0.649			
34		Radius r Quadrant	mm			
35		Orifice Bore Diameter	mm	284.4		
36		Vent/Drain Hole Requirement	Vent			
37		Vent/Drain Hole Dia.	mm	10.32		
38		Centre Distance, R	mm			
39		Plate Thickness, W	mm	9.52		
40		Reynolds Number	524542			
ORIFICE FLANGES	41	Flanges	Not Required			
	42	Type of Flange				
	43	Tap Size	mm			
	44	No. of Taps per flange				
	45	Flange Material				
	46	Size & Rating				
	47	Facing & Finish				
	48	Gasket thickness	mm			
	49	Gasket Material				
	50	Stud Bolt Material				
	51	Nut Material				
CERTIFICATION	52	Statutory	NA			
	53	Others	NA			
OPTIONS	54	Manufacturers	*			
	55	Model No.	*			
NOTES 1. Orifice assembly must include supply of Plate. 2. Gasket Thickness: As per B-16.20- ANSI B16.5. 3. Allowable pressure drop: 0.3 Kg/cm2g. 4. This tag is D-D/2 type, so orifice flanges are in purchaser's scope.						
[] Deviation		[] No Deviation		Vendor Seal		
0		10-Feb-2025		Issued For Bids		
Rev. No.		Date		Purpose		
Prepared By		Checked By		Approved By		